

Rahul Gupta

Senior Data Science, Machine Learning, and Responsible AI Leader

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EXECUTIVE PROFILE

- Senior applied science and data science leader in Amazon's AGI organization, driving AI strategy, evaluation, responsible AI, model quality, and operational mechanisms for frontier and large-scale ML systems.
- Built and led technical organizations of 50+ across Amazon Nova safety, LMArena performance, and Responsible AI programs; experienced managing scientists, technical leaders, and cross-functional programs across product, engineering, research, and executive stakeholders.
- Hands-on scientist with first-author work on [CBRN uplift evaluation for frontier language models](#) and [fairness in large-scale NLU systems](#); 25+ patented or patent-pending technologies and publications across NeurIPS, ICLR, ACL, EMNLP, FAccT, and ICASSP.

DATA SCIENCE & MACHINE LEARNING LEADERSHIP

Senior Applied Science Manager

June 2026 – Present

Amazon / Frontier AI Assets, Responsible AI

Cambridge, MA

- Overall head of Responsible AI in Amazon's newly architected AGI division, reporting directly to a Vice President.
- Set research and execution priorities for frontier model safety, evaluation, red teaming, fairness, privacy, policy alignment, and launch readiness.
- Translate ambiguous AI risks into measurable frameworks, model-quality signals, executive guidance, and scalable mechanisms for product and technology teams.

Senior Applied Science Manager

October 2022 – May 2026

Amazon AGI / Nova Responsible AI

Cambridge, MA

- Led Nova Responsible AI programs covering safety evaluations, red teaming, fairness, privacy-sensitive model behavior, and mitigation strategies for advanced AI systems.
- Led the Nova LMArena team, driving Nova models to top-15 LMArena performance by connecting evaluation diagnostics, post-training priorities, and model iteration.
- Founded and led the Nova Trusted AI Challenge, generating 100+ applications and selecting 10 university teams to advance external safety and evaluation research.
- Built governance and reporting mechanisms used with senior leaders and external evaluators to assess model quality, risk, readiness, and research progress.

Applied Science Manager

December 2018 – September 2022

Amazon Alexa / Responsible AI

Cambridge, MA

- Founded and led a Responsible AI research team for Alexa, building foundational capabilities for private, fair, and trustworthy customer experiences across platform and product workflows.
- Partnered with product, engineering, and senior executive stakeholders to operationalize responsible AI goals and report outcomes to Amazon Senior Vice Presidents.
- Helped embed responsible AI mechanisms into large-scale conversational AI systems used by broad consumer populations.

Applied Scientist

August 2016 – November 2018

Amazon Alexa / Large-Scale NLU

Cambridge, MA

- Served as technical lead on decoupling training and testing of large-scale ML systems, improving evaluation rigor for production NLU development.
- Delivered business impact that led to organizational restructuring and influenced production model development practices.

EDUCATION

PhD, *Electrical Engineering, University of Southern California*

2010 – 2016

Bachelor of Technology, *Electrical Engineering, Indian Institute of Technology, Kharagpur*

2006 – 2010

RESEARCH LEADERSHIP & SERVICE

- Organizer of the [TrustNLP](#) workshop series and workshops on trustworthy speech processing and LLM unlearning [1], [2]; reviewer for ACL, ICLR, ICASSP, and related top-tier venues.
- Point of contact for multiple [Amazon Academic Research Awards](#) collaborations, collectively representing over \$20M in funded research.
- Keynote speaker at venues including [IEEE SLT](#) and workshops on frontier AI advancement and social impact [1], [2].

SELECTED PUBLICATIONS

Full list available at my [Google Scholar](#) page.

- [A Threshold Exceedance Framework for CBRN Uplift Evaluation in Frontier Language Models](#), arXiv preprint, 2026. First-author, hands-on experimentation.
- [Establishing Best Practices for Building Rigorous Agentic Benchmarks](#), NeurIPS 2025. Winner of the AgentX LLM Agents MOOC Competition, 2025.
- [FLIRT: Feedback Loop In-context Red Teaming](#), EMNLP 2024. Covered in the red teaming community [1], [2]; #3 paper of the day on Hugging Face.
- [“I’m fully who I am”: Towards centering transgender and non-binary voices to measure biases in open language generation](#), ACM FAccT 2023.
- [BOLD: Dataset and Metrics for Measuring Biases in Open-Ended Language Generation](#), ACM FAccT 2021. (700+ citations)

OTHER RECOGNITIONS

- Ming Hsieh Department of Electrical Engineering best PhD pitch award, 2015.
- Two-time winner of Interspeech computational paralinguistics challenge, 2013 and 2015.
- Phi Beta Kappa alumni in Southern California scholarship, 2015.
- Ming Hsieh Department of Electrical Engineering humanitarian award, 2014.
- Provost PhD fellowship, 2010–2014.